Logarithmic biorthogonal norms for the Gaussian Gabor system

MathDB 375037

Principia Math in Progress

12 September 2026

Task 1. Prove Conjecture 4.1 of Zikkos [1]: after deleting the Gaussian Gabor atom indexed by $(0,0)$, the unique biorthogonal vectors satisfy

$$\|\gamma_{m,n,0,0}\|^2_{L^2(\mathbb{R})} \asymp \log\Big(e+\sqrt{m^2+n^2}\Big), \qquad (m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}. \tag{1}$$

Also prove the stated consequence for positive weights whose logarithmic squared-reciprocal sum is finite. All comparison constants below are independent of (m,n).

Theorem 1. *Estimate* (1) *holds. If* $c_{m,n}>0$ *and*

$$\sum_{(m,n)\neq(0,0)} \frac{\log(e+\sqrt{m^2+n^2})}{c_{m,n}^2} < \infty, \tag{2}$$

then the family

$$\Big\{c_{m,n}e^{2\pi int}e^{-\pi(t-m)^2} : (m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}\Big\}$$

is a complete Riesz–Fischer sequence and a lower semi-frame for $L^2(\mathbb{R})$.

The Zak transform and its zero. Initially normalize the Gaussian as $\phi(t)=2^{1/4}e^{-\pi t^2}$. The Zak transform

$$Zf(x,\xi)=\sum_{j\in\mathbb{Z}} f(x-j)e^{2\pi ij\xi}$$

is unitary from $L^2(\mathbb{R})$ onto $L^2(Q)$, where $Q=[0,1)^2$. Write $\Theta=Z\phi$ and $E_{m,n}(x,\xi)=e^{2\pi i(nx-m\xi)}$. Changing the summation index gives

$$Z\big(e^{2\pi int}\phi(t-m)\big)=E_{m,n}\Theta. \tag{3}$$

The classical theta-function description gives a unique zero of Θ in Q, at $q_0=(1/2,1/2)$. It is a nondegenerate zero:

$$c_0|u|\le|\Theta(q_0+u)|\le C_0|u| \quad (|u|<\delta) \tag{4}$$

for positive c_0,C_0 and $0<\delta<1/2$. For clarity, the published proof in [2, Corollary 4.3, equation (11)] gives the expansion

$$\Theta(q_0+u)=-2^{1/4}\pi\theta_1'(0,e^{-\pi})(u_1+iu_2)+O(|u|^2), \qquad \theta_1'(0,e^{-\pi})\neq 0.$$

The derivative is nonzero also by the Jacobi triple product. The modulus of Θ is periodic in both coordinates and bounded away from zero on the complement of this disk in the closed unit square. In particular, $1/\Theta\notin L^2(Q)$.

The biorthogonal formula. For $(m,n) \neq (0,0)$ let

$$a_{m,n} = E_{m,n}(q_0) = (-1)^{n-m}, \qquad H_{m,n} = \frac{E_{m,n} - a_{m,n}}{\overline{\Theta}}.$$

The numerator vanishes to at least first order at q_0. By (4), this quotient is locally bounded for each fixed index, and it is bounded away from the zero as well. Thus $H_{m,n} \in L^2(Q)$. Let $h_{m,n} = Z^{-1}H_{m,n}$. For all nonzero (r,s), Fourier orthogonality gives

$$\begin{aligned}\left\langle e^{2\pi ist}\phi(t-r), h_{m,n}\right\rangle &= \int_Q E_{r,s}\Theta\,\frac{\overline{E_{m,n} - a_{m,n}}}{\overline{\Theta}}\,dA \\ &= \int_Q E_{r,s}\overline{E_{m,n}}\,dA - \overline{a_{m,n}}\int_Q E_{r,s}\,dA = \delta_{r,m}\delta_{s,n}. \end{aligned} \tag{5}$$

The constant term disappears because the origin index was deleted.

For completeness, if $F \in L^2(Q)$ is orthogonal to all $E_{r,s}\Theta$ with $(r,s) \neq (0,0)$, the function $F\overline{\Theta}$ has only a constant Fourier coefficient. It is therefore constant almost everywhere. Since $1/\Theta \notin L^2(Q)$, the constant must be zero, and $F = 0$ almost everywhere. Thus the deleted system is complete. The exhibited biorthogonal vectors imply minimality, and completeness implies uniqueness of the biorthogonal family. This recovers the known formula in [2, Theorem 4.2]. In particular,

$$\|h_{m,n}\|_2^2 = \int_Q \frac{|E_{m,n} - a_{m,n}|^2}{|\Theta|^2}\,dA. \tag{6}$$

Lemma 1. *For fixed $\delta > 0$, define*

$$J_\delta(p) = \int_{|u|<\delta} \frac{|e^{2\pi ip\cdot u} - 1|^2}{|u|^2}\,dA(u).$$

As $|p| \to \infty$, $J_\delta(p) \asymp \log|p|$, uniformly in the direction of $p \in \mathbb{R}^2$.

Proof. Rotate polar coordinates so that p points along the first axis and put $R = |p|$. With $J_0(s) = (2\pi)^{-1}\int_0^{2\pi}\cos(s\cos\theta)\,d\theta$, the angular integral gives

$$J_\delta(p) = 4\pi\int_0^{2\pi\delta R}\frac{1 - J_0(s)}{s}\,ds. \tag{7}$$

The inequality $1 - \cos t \leq t^2/2$ implies $0 \leq 1 - J_0(s) \leq s^2/4$ near zero. For every $s > 0$, $J_0(s) < 1$, since the cosine in its angular average is not identically one. Moreover,

$$J_0(s) = \frac{1}{\pi}\int_{-1}^{1}\frac{\cos(st)}{\sqrt{1-t^2}}\,dt \longrightarrow 0$$

by the Riemann–Lebesgue lemma. Continuity and compactness therefore give a fixed $\eta > 0$ such that $\eta \leq 1 - J_0(s) \leq 2$ for every $s \geq 1$. The integral over $(0,1)$ in (7) is bounded, while that over $(1, 2\pi\delta R)$ is comparable to $\log R$. The formula depends only on R, giving uniformity in direction. □

Completion of the estimate. In the disk around q_0,

$$E_{m,n}(q_0+u) = a_{m,n}e^{2\pi i(nu_1-mu_2)}.$$

Equations (4) and (6) sandwich the local norm contribution between fixed positive multiples of $J_\delta((n,-m))$. On the complement, the numerator is at most two in modulus and $|\Theta|$ is bounded below, so the remaining contribution is bounded uniformly in (m,n). Lemma 1 gives $\|h_{m,n}\|_2^2 \asymp \log\sqrt{m^2+n^2}$ for large radius. There are only finitely many nonzero lattice points in a bounded disk, and each corresponding norm is positive. Adjusting the constants therefore proves the uniform bound with $\log(e+\sqrt{m^2+n^2})$.

The source uses $e^{-\pi t^2} = 2^{-1/4}\phi(t)$, so its biorthogonal vectors are $\gamma_{m,n,0,0} = 2^{1/4}h_{m,n}$. Their squared norms differ by the fixed factor $\sqrt{2}$. This proves (1).

Finally, (2) implies $\sum \|\gamma_{m,n,0,0}\|_2^2/c_{m,n}^2 < \infty$. The deleted system is complete and minimal, as proved above. Theorem 1.2 of [1] now gives both the Riesz–Fischer and lower semi-frame assertions, including completeness. □

Prior results and verification. The explicit biorthogonal formula and the simple Zak zero are known. Belov [3, equation (3.1)] already proved the logarithmic upper bound in Fock coordinates; [4, Lemma 2.3] repeats it. The argument here supplies the uniform matching lower bound and the complete two-sided estimate requested in the 2026 conjecture. Exact Fourier checks verify the index and phase conventions; supplementary numerical checks compare the Gaussian Zak series with the theta formula and independently check the angular integral. Finite computations are not used to infer the asymptotic statement.

Task 2. Determine the leading constant in the logarithmic norm asymptotic and prove a uniform bounded remainder using the local Taylor expansion of the Gaussian Zak transform.

Task 3. Find necessary and sufficient growth conditions for radial weights $c_{m,n} = c(\sqrt{m^2+n^2})$ to produce a lower semi-frame, beyond the sufficient summability condition (2).

Task 4. Extend the norm-growth analysis to other windows whose Zak transforms have isolated zeros, including higher-order zeros and the deletion of an appropriate finite set of atoms.

Task 5. Obtain quantitative stability bounds for finite reconstruction from the weighted Gaussian system, relating the truncation radius and weight growth to the norms of the biorthogonal vectors.

References

[1] E. Zikkos, *The Gaussian Gabor system at the critical density is a weighted lower semi frame*, 2026. arXiv:2606.00764, Conjecture 4.1.

[2] L. Hart, C. Heil, I. Katz, and M. Northington V, *Overcomplete reproducing pairs*, J. Fourier Anal. Appl. **31** (2025), article 58. Published text.

[3] Y. Belov, *Uniqueness of Gabor series*, Appl. Comput. Harmon. Anal. **39** (2015), 545–551. arXiv:1409.5231.

[4] A. Baranov, Y. Belov, and A. Borichev, *Summability properties of Gabor expansions*, J. Funct. Anal. **274** (2018), 2532–2552. arXiv:1706.05685.